MINDX · START PROGRAMME

Technical foundations for multiuser XR enhanced neuromodulation in digital healthcare

Hyperscanning neurofeedback with fNIRS and mixed reality

Status report with a demonstration on synthetic data

PROBLEM STATEMENT

Social synchrony is measurable, but not yet trainable

- Social interaction shapes brain development and mental health
- Interpersonal neural synchrony (INS) is only associated with outcomes, not shown to be causal
- Neurofeedback trains one person's signal amplitude
- Hyper-NF is nascent: 9 of 120 studies, one fNIRS study, a single dyad

HYPOTHESES

Three testable claims

ID	CLAIM	EVIDENCE
H1	Dyads raise INS with joint feedback	Early versus late sessions, controlled for baseline
H2	The effect exceeds sham feedback	Within group, one sham session in three
H3	Cooperation transfers without feedback	Exchange game, pre versus post

SOLUTION APPROACH

A closed loop around the dyad

Measure
Two NIRSport2 systems, PFC and right TPJ, short channels

Compute
Causal online processing and one joint INS value

Feed back
Speed and pitch of a shared car in mixed reality

Evaluate
Regulation, sham control, transfer task

All streams share one LSL clock. Storage runs off the feedback path.

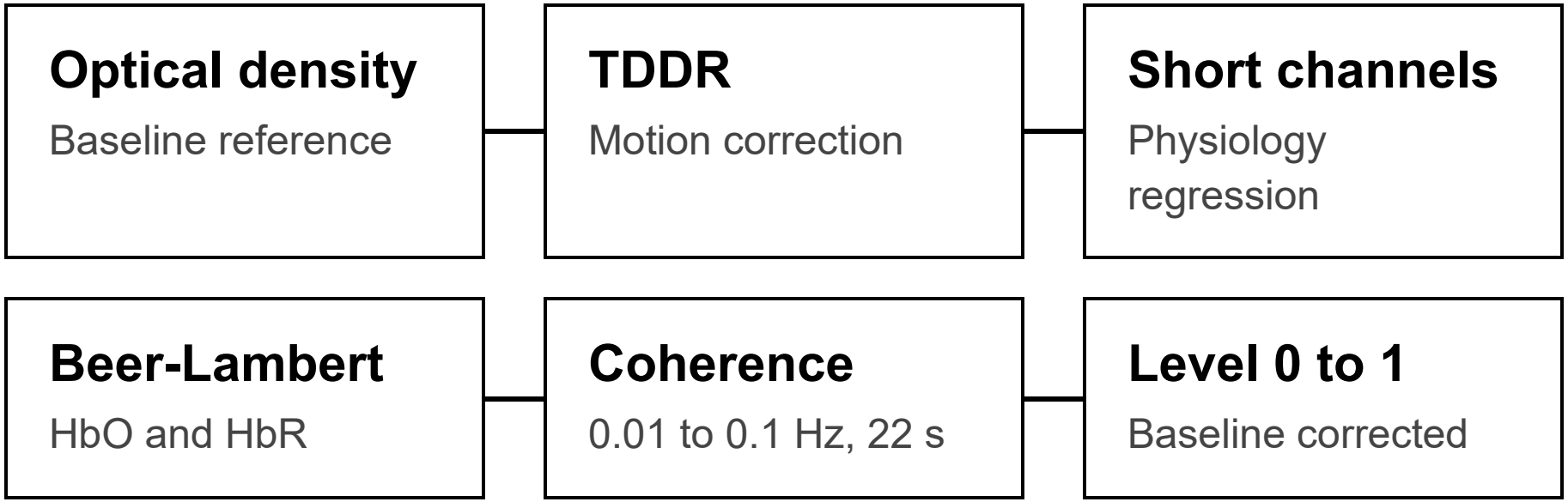
DESIGN RATIONALE

Why these choices

CHOICE	REASON
fNIRS	Better localisation than EEG, tolerates motion, suits adolescents
Joint INS value	Targets the social process itself, not one brain
Causal processing	No future samples, so feedback is never late
Mixed reality	Partners see each other, which favours transfer
Sham from another dyad	Controls reward and unrelated training effects

SIGNAL CHAIN

From photons to a feedback level



Implemented with MNE-NIRS, causal at every step.

RESULTS SO FAR

The estimator follows injected synchrony

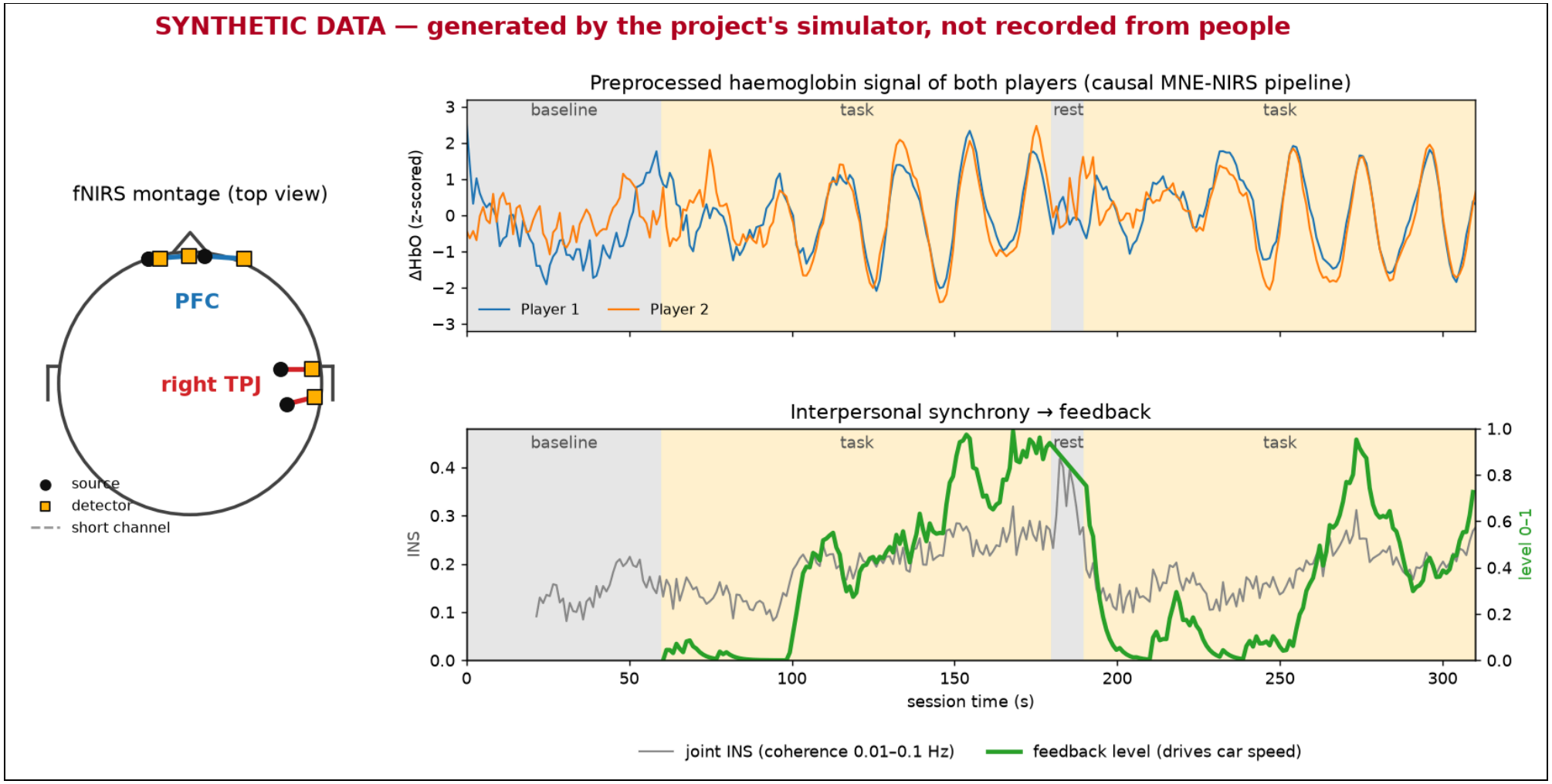
0.08 to 0.17
INS when injected coherence rises from 0.1 to 0.9

22 s
window: a 60 s window saturates and loses range

7 ms
mean processing time per frame

- Monotonic response on synthetic dyads, with and without the MNE montage
- More than 110 automated tests on backend and game

Synthetic fNIRS through the real pipeline



RESULTS SO FAR · GAME

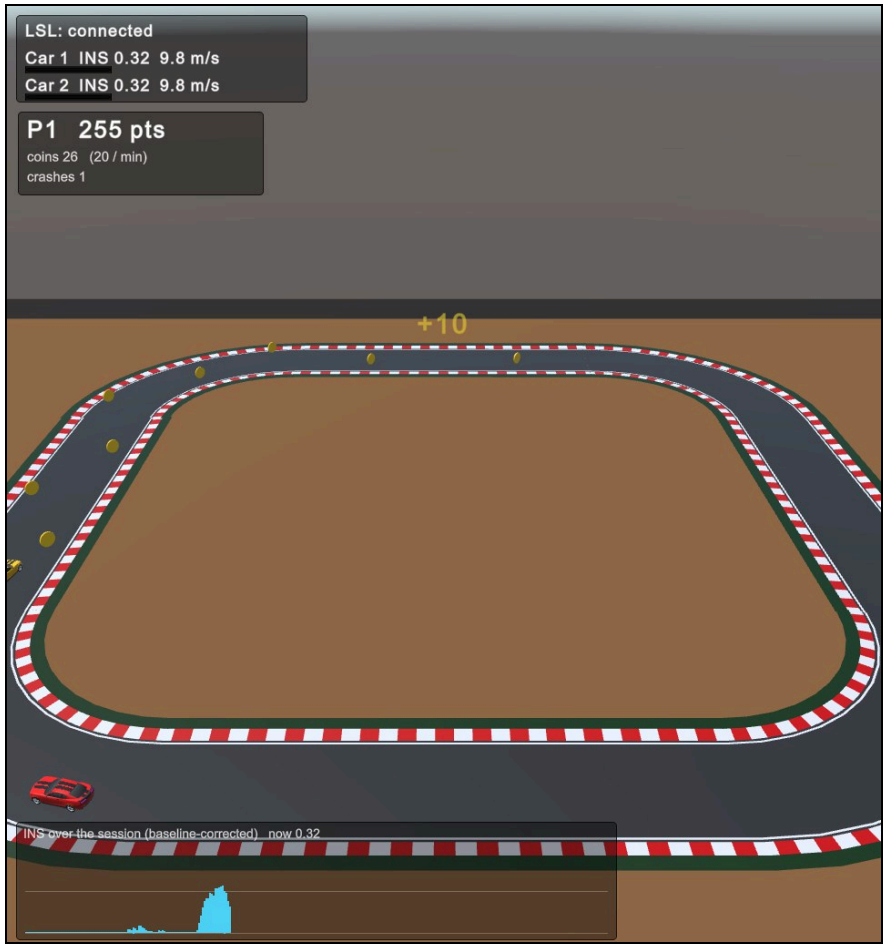
Baseline: level zero



- Speed stays at 5 m/s
- Own score in each player's view
- Coins pay 10 points

RESULTS SO FAR · GAME

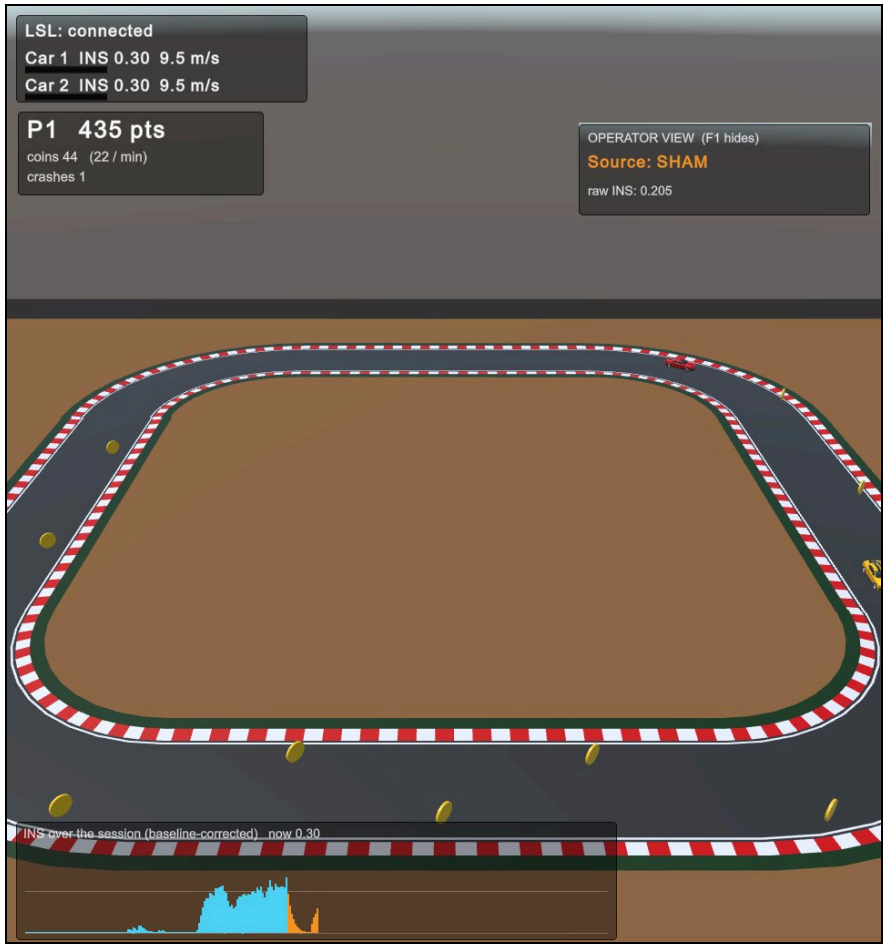
Synchrony rises, the car speeds up



- Speed from 5 to 17 m/s
- Coins per minute from 11 to 27
- Delay of about 30 s, set by physiology

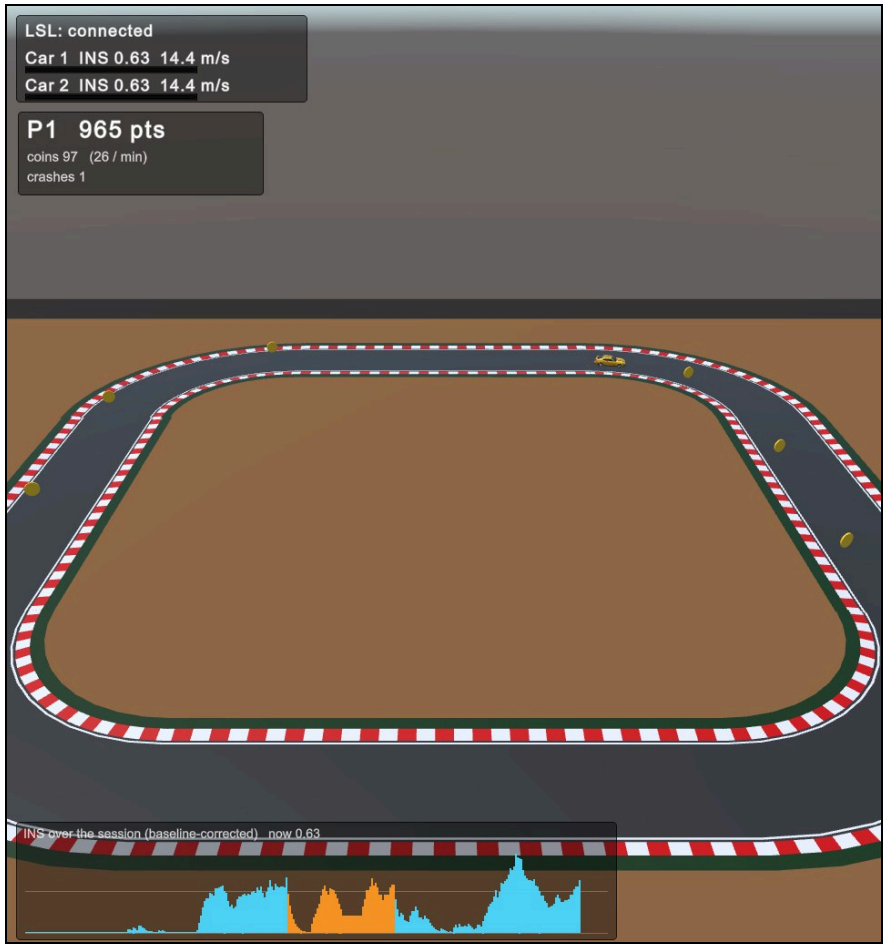
RESULTS SO FAR · CONTROL

Sham feedback, invisible to players



- Another dyad's recording, same mapping
- Operator switch, no visible change
- The sham trace also rises: it is a real trajectory

Reward and penalty per player



- Plus 10 per coin in the player's own lane
- Minus 5 for both cars on a crash
- Scores differ by lap length, about 7%

RESULTS SO FAR · PLACEHOLDER

One signal per player

PLAYER	LEVEL	SPEED	COINS PER MINUTE
Player 1	0.80	17.1 m/s	38
Player 2	0.39	10.8 m/s	22

Demo placeholder feature: own HbO band power. It does not settle the open choice of a per-participant feature.

RESULTS AND TARGETS

Where we stand, what we need

Obtained

- Closed loop, end to end
- Causal MNE-NIRS pipeline
- Sham mechanism
- Game with reward and penalty
- Simulated table view

To obtain

- SUS above 68 percent
- PSSUQ of 3 or lower
- Regulation: early versus late session
- Effect versus sham
- Transfer in the exchange game

CHALLENGES

What still has to be solved

- Live NIRSport2 input over LSL with signal quality checks
- INS window and estimator: stability against latency
- A per-participant feature for Individual mode
- Reward design: items reachable only at a held velocity
- Co-located multiplayer XR on the real headset

OUTLOOK

From synthetic to two people live

STEP	CONTENT	TIME
Level B	One person, live fNIRS	2 to 4 weeks
Level C	Two people, live and co-located	1 to 2 months or more
Study	Three sessions, one sham, transfer task	After ethics approval
Analysis	Bayesian models, open source toolbox	WP5